

Scheduling (reliable triggers)

Audience: operators / engineers

Chinese version: SCHEDULING.zh-TW.md (./SCHEDULING.zh-TW.md)

GitHub's native schedule cron is often delayed or skipped. This project uses:

1. Primary: an external cron service calling the GitHub API (workflow_dispatch)
 2. Backup: sparse GitHub cron (Taiwan 10:30 / 15:30 / 18:30, weekdays)
 3. Manual: Actions UI or local scripts/trigger_update.sh
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1. Recommended external cron (cron-job.org, free)

1.1 Create a GitHub token

1. GitHub → Settings → Developer settings → Personal access tokens
2. Prefer a fine-grained token:
 - Repository access: only DownloadBankOfTaiwanExchangeRate
 - Permissions: Actions: Read and write
1. Copy the token (shown once)

Never commit the token. Store it only in the external cron service's secret fields.

1.2 Create a job on cron-job.org

1. Sign up at <https://cron-job.org> (<https://cron-job.org>)
2. Create a cronjob:

Field	Value
Title	BOT FX Rates - Update
URL	https://api.github.com/repos/ChesleyLo/DownloadBankOfTaiwanExchangeRate/actions/workflows/update-rates.yml/dispatch
Schedule	See below
Request method	POST
Timezone	Asia/Taipei

Headers:

```
Accept: application/vnd.github+json
Authorization: Bearer <YOUR_GITHUB_TOKEN>
X-GitHub-API-Version: 2022-11-28
Content-Type: application/json
```

Body:

```
{"ref": "main"}
```

1.3 Suggested times (Taiwan)

09:10, 10:10, 11:10, 12:10, 13:10, 14:10, 15:10, 16:10, 18:30

Custom cron (if supported), timezone Asia/Taipei:

```
10 9-16 * * 1-5
30 18 * * 1-5
```

1.4 Success signals

- cron-job.org shows HTTP 204
- GitHub Actions shows a workflow_dispatch run for Update BOT FX Rates

2. Local / server trigger

```
# A) gh already logged in
./scripts/trigger_update.sh

# B) PAT
export GITHUB_TOKEN=ghp_xxx
./scripts/trigger_update.sh

# C) repository_dispatch
./scripts/trigger_update.sh --mode repository_dispatch
```

Example crontab:

```
10 9-16 * * 1-5 GITHUB_TOKEN=ghp_xxx /path/to/repo/scripts/trigger_update.sh >>/tmp/bot-fx.log 2>&1
30 18 * * 1-5 GITHUB_TOKEN=ghp_xxx /path/to/repo/scripts/trigger_update.sh >>/tmp/bot-fx.log 2>&1
```

3. GitHub backup schedule

UTC	Taiwan	Role
30 2 * * 1-5	10:30	backup
30 7 * * 1-5	15:30	backup
30 10 * * 1-5	18:30	backup

Treat the external cron as primary.

concurrency.group: update-bot-rates prevents overlapping runs from fighting each other.

4. Troubleshooting

Symptom	Action
401/403 from cron	Token missing Actions: write, or expired
404	Wrong workflow file name / repo path
204 but no Actions run	Workflow disabled, or wrong ref
Run OK but no commit	Rates unchanged (changed=false) — normal

5. Related files

- `.github/workflows/update-rates.yml`
- `scripts/trigger_update.sh`
- `TECHNICAL.en.md` (`./TECHNICAL.en.md`)